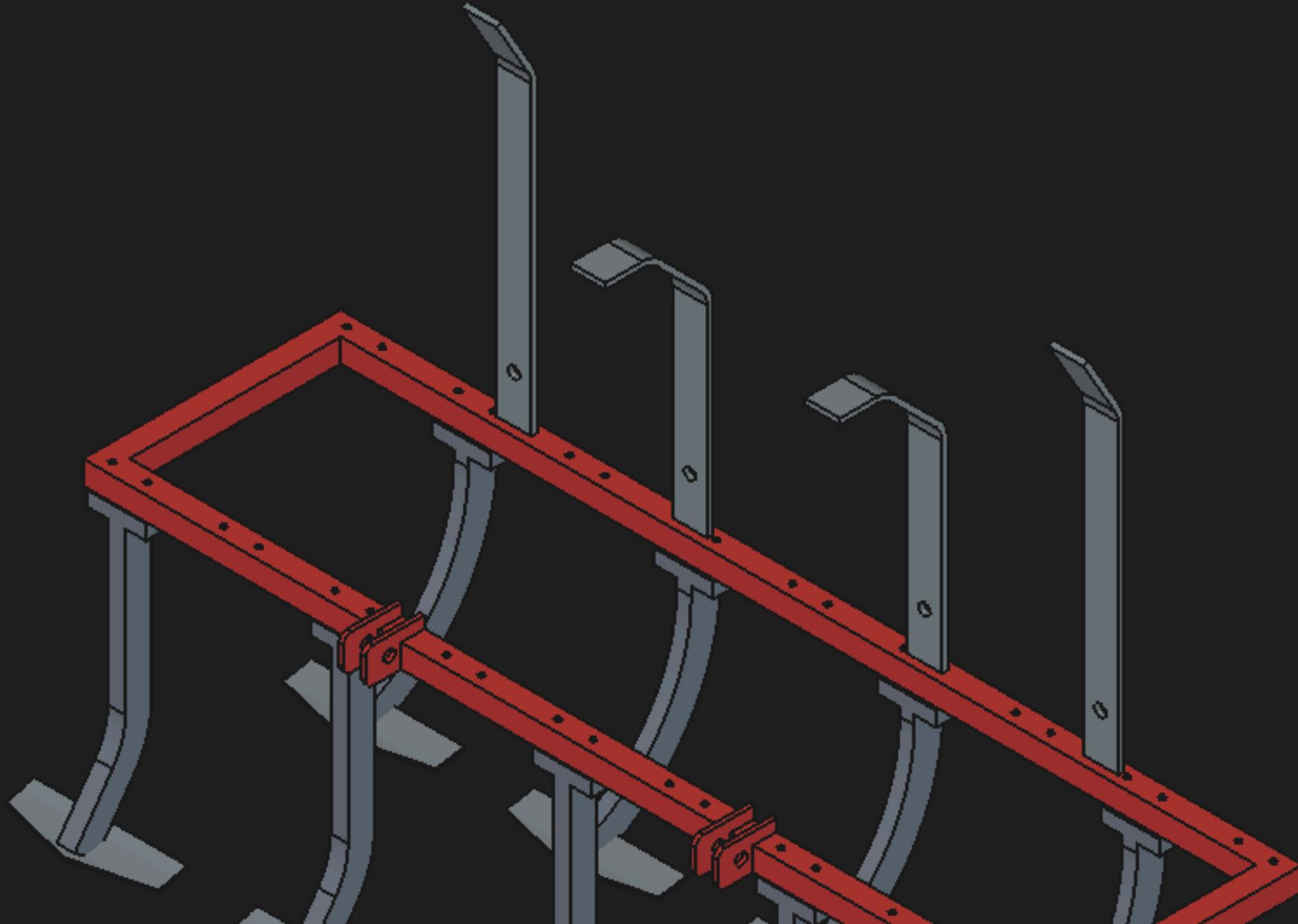
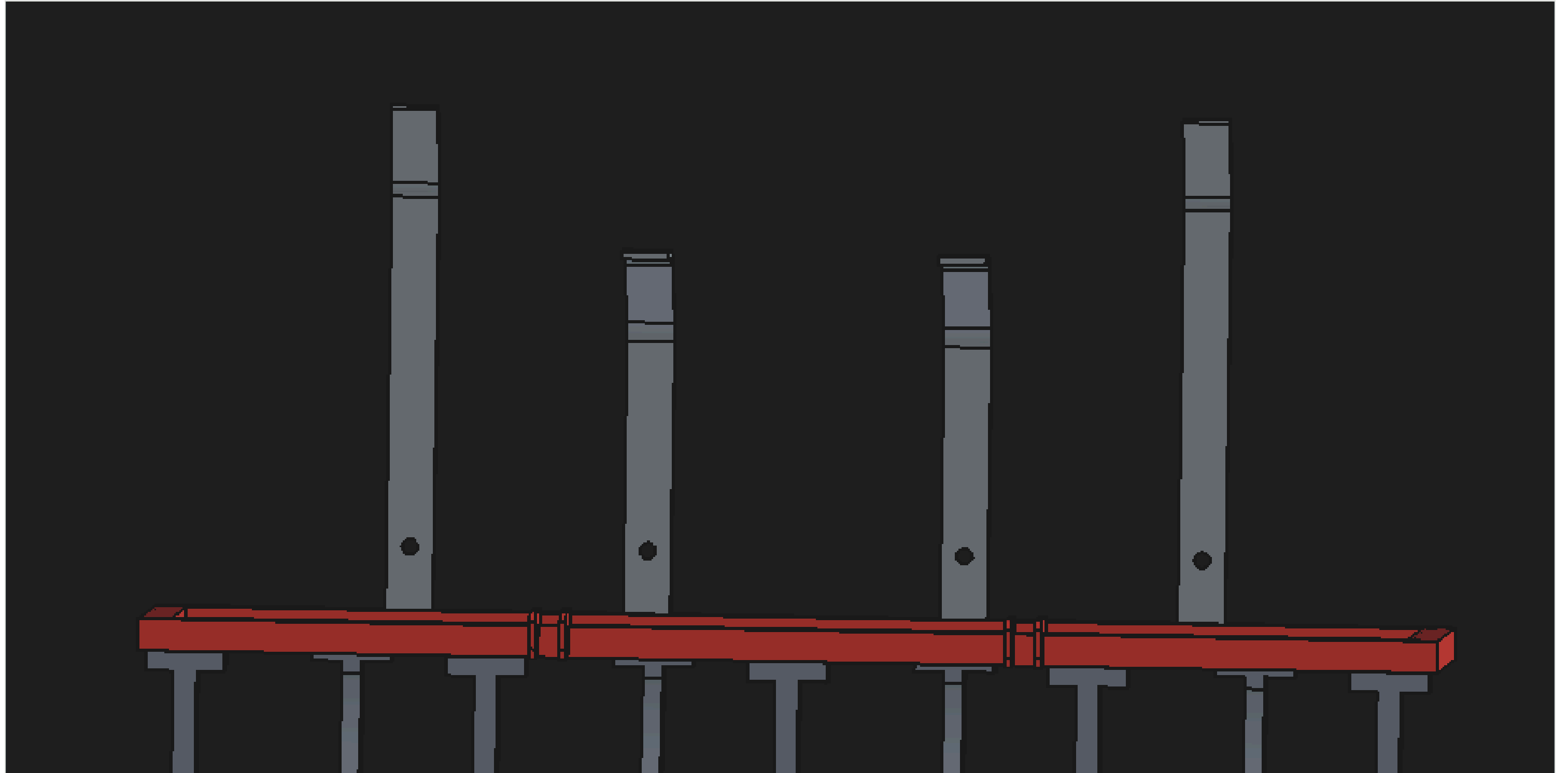


The Nine-Tyne Cultivator



What a cultivator does



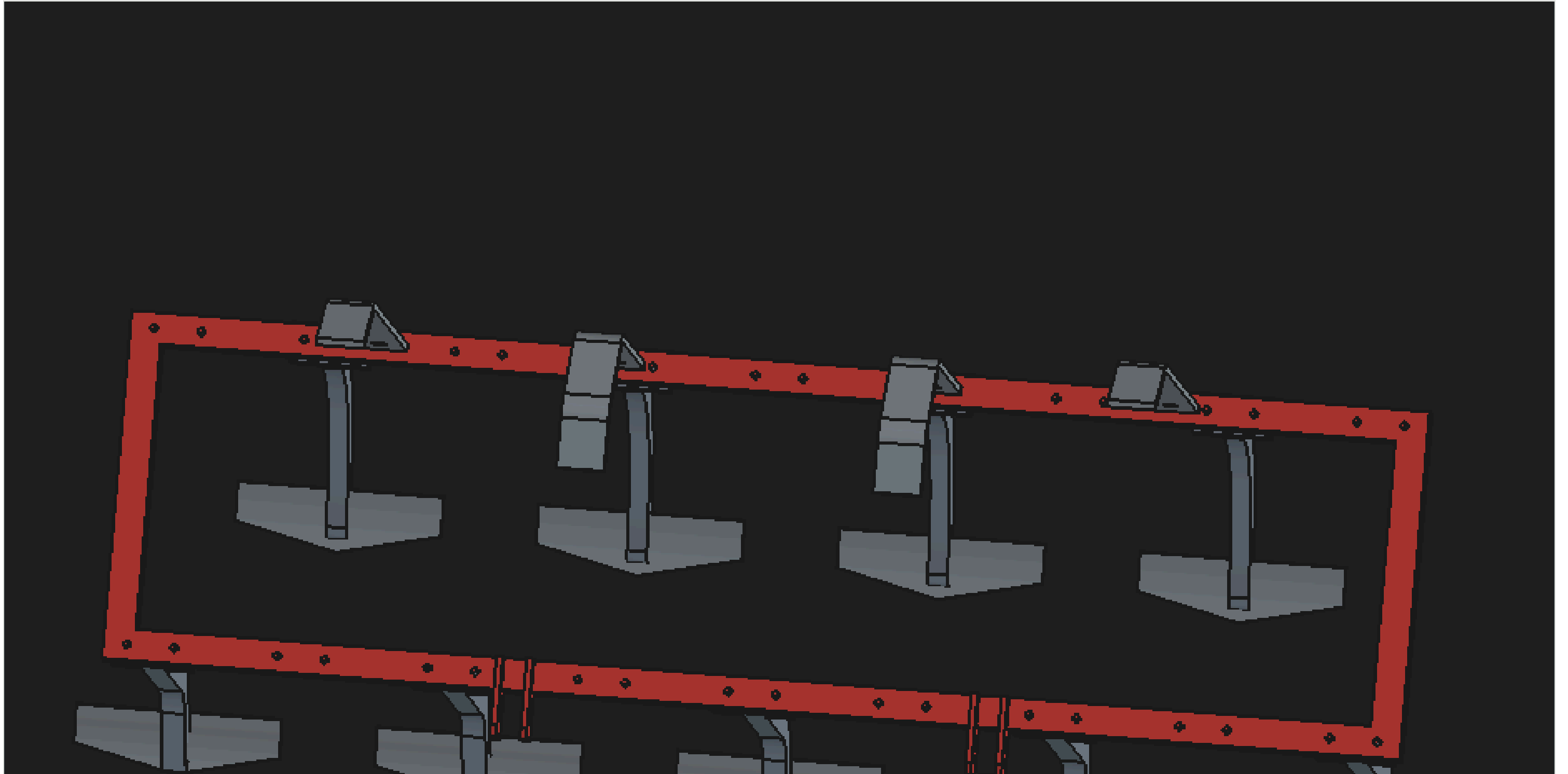
Where it fits in the field work

Every tillage tool has its place in the sequence. The cultivator is the second pass.

PASS	TOOL	WORKING DEPTH
First	Mouldboard or disc plough	8 to 12 inches
Second	Cultivator — this machine	4 to 9 inches
Third	Disc harrow or tine harrow	2 to 4 inches
Last	Roller or planker	surface only

Nine tynes at 9.5 inch spacing cover 76 inches in one pass.
That suits a 35 to 45 horsepower tractor.

Why the tynes are in two rows

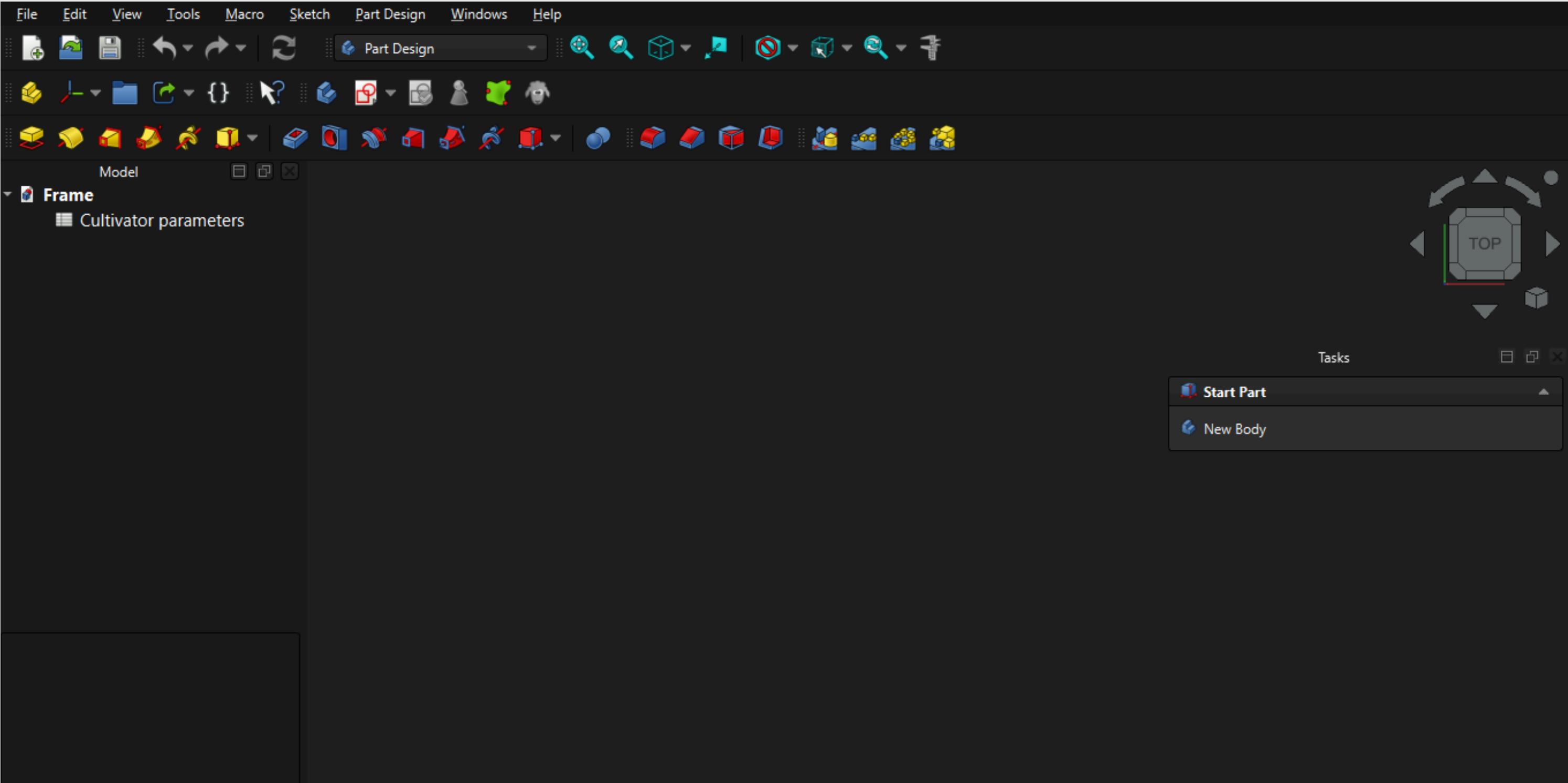


The seven parts

Seven different parts make up the machine. You will model every one.

PART	HOW MANY	WHAT IT DOES
1. Frame	1	The welded square-tube frame that carries everything
2. Tine 01	5	A straight tyne with a bent foot, for the front row
3. Tine 02	4	A curved tyne, for the back row
4 & 5. Clamps	2 + 2	Bent steel straps that brace the frame
6 & 7. Blades	5 + 4	The sweeps that do the cutting

A model, not a drawing



Where the design comes from

This course follows a SolidWorks tutorial by Malviya CAD Solution. The original is in Hindi, and we worked from a rough translation.

- Some numbers in the translation are garbled. One line calls a hole “12.5”, another calls it “3 inches”.
- Neither is a diameter. The 3 inches turned out to be the *spacing* between a pair of holes.
- So we read the tutorial, check it against what the machine has to do, and then decide.

Wherever we had to decide something the source did not say, the slide says so. Deck 14 lists all twelve of those decisions in one place.

How the course is laid out

LEARNING FREECAD

DECK	
2	The FreeCAD window
3	Setting up a part
4	Sketching and constraints

BUILDING THE PARTS

DECK	
5	The frame
6	Tine 01
7	Tine 02
8	The clamps
9	The blades

PUTTING IT TOGETHER

DECK	
10	The assembly
11	Patterns and fasteners
12	Appearance and drawings
13	Making it parametric
14	Summary and exercises

You need FreeCAD 1.1 or newer. It is free. Nothing in this course needs a paid add-on.

Allow about twelve hours for the whole thing.